Components, Ohm's law & resistor networks

ENGINEERING WITH MR KELLY

National 5 · Electronics & control · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ recognise and draw standard component symbols
- ☐ use Ohm's law and the power formulae
- ☐ calculate total resistance of series and parallel resistors
- ☐ show where an ammeter (series) and voltmeter (parallel) are connected
- ☐ describe advantages of simulating a circuit before building it

I DO — Worked example

Read how a full-mark answer is built

A pupil wants to simulate a circuit before building it.

1 Explain one advantage of simulating the circuit before constructing it. (2 marks)

Full-mark answer each ✓ is one mark

Faults in the circuit can be found and corrected on screen before it is built ✓, so no components are damaged or wasted ✓.

Why marks are not awarded

"It is cheaper and it is quicker."

0 marks: two vague benefits, no cause and effect.

WE DO — Guided practice

Use the starters and word bank to help you

Two 470 Ω resistors are connected in parallel.

2 Calculate the total resistance. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Two 1 kΩ resistors in parallel have a total resistance of…
☐ 2 kΩ ☐ 1 kΩ ☐ 250 Ω ☐ 500 Ω

b) In a series circuit with 9 V supply, V1 = 3.5 V and V2 = 2.5 V. V3 is…
☐ 3 V ☐ 6 V ☐ 15 V ☐ 2.5 V

c) 12 V across 680 Ω gives a current of about…
☐ 0.018 A ☐ 56 A ☐ 8160 A ☐ 1.8 A

d) UK mains electricity is…
☐ 12 V d.c. ☐ 230 V d.c. ☐ 230 V a.c. ☐ 110 V d.c.

4 Key terms: write the letter of the matching description next to each term.

1. **Conventional current**	____	**A** R = (Vs − V LED) ÷ I
2. **Kirchhoff's first law**	____	**B** Flows from + to − (electrons flow the other way).
3. **Kirchhoff's second law**	____	**C** No current until the switch is operated (push-to-make).
4. **LED series resistor**	____	**D** Limit switch with C, NO and NC terminals; detects position.
5. **Microswitch**	____	**E** Voltages round a series loop add up to the supply.
6. **Normally open (NO)**	____	**F** Currents into a junction = currents out: IT = I1 + I2 + …

A 12 V lamp draws 0.5 A.

5 Calculate the power of the lamp. (2 marks)

An ammeter is needed to measure the current through a lamp.

6 Draw the symbol for an ammeter and state how it is connected. (2 marks)

In a series circuit with R1 and R2, R1 is increased.

7 State the effect on V1 and on the current. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?